EDS Practical 05

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5.1.1 Stacked plot

Code: import matplotlib.pyplot as plt import pandas

as pd

#

Data for Months and Temperature for three cities data = { 'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December'],

'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18,

12, 8, 6],

'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9,

5, 3],

'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7,

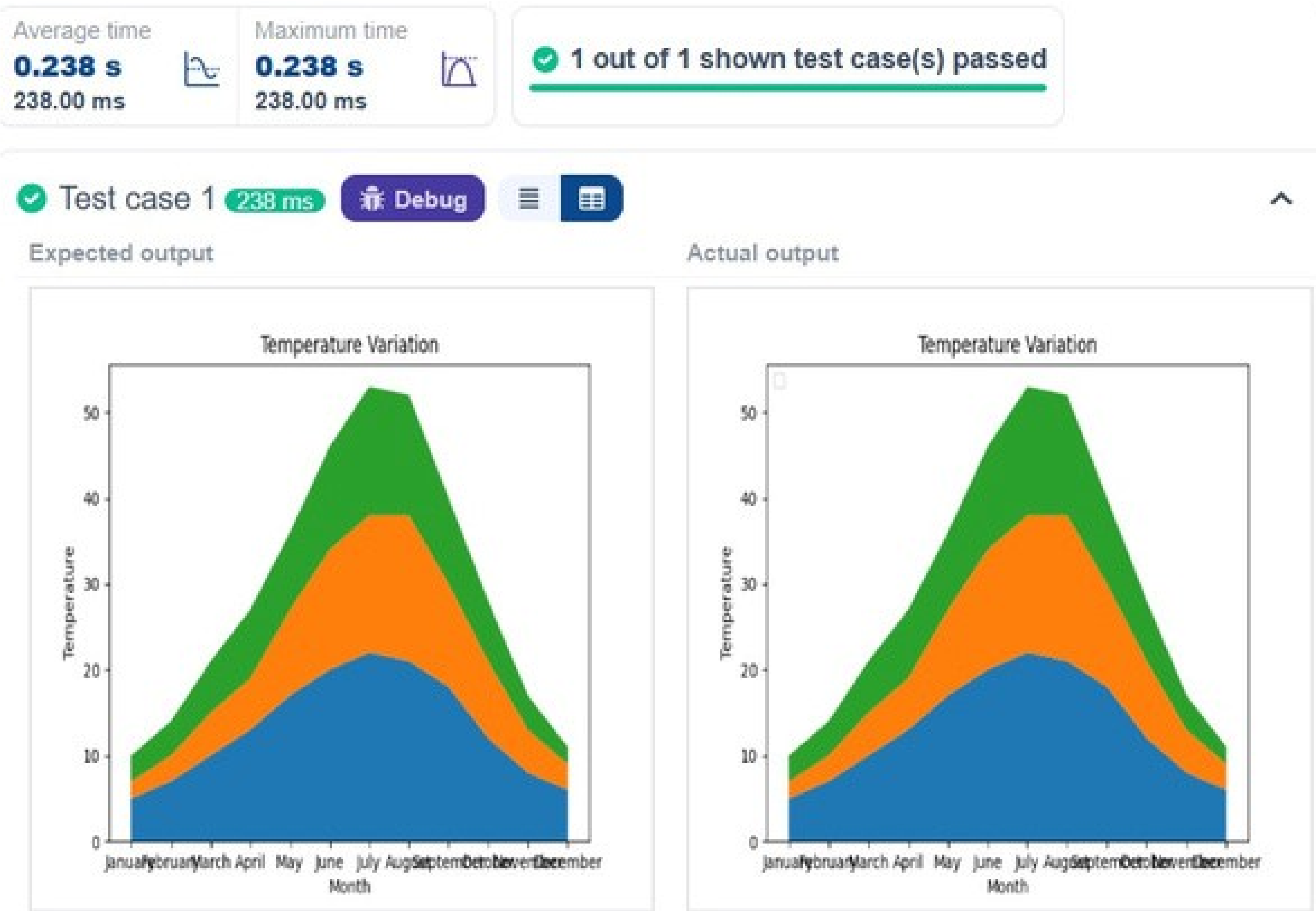
4, 2] }

#

Write your code...

df = pd.DataFrame(data)

plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_
B_Temperature'],df['City_C_Temperature']) plt.xlabel('Month')
plt.ylabel('Temperature') plt.title('Temperature
Variation') plt.legend(loc='upper left') plt.show()



1. Titanic Dataset Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset from the CSV file df = pd.read_csv('titanic.csv')

#

Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))
#

write the code..

##

plot 1:count of passengers by class

axes[0, 0].bar(df['Pclass'].value_counts().index,df['Pclass'].value_ counts(), color='skyblue') axes[0, 0].set_title("Passenger Class Distribution") axes[0,0].set_xlabel("Pclass") axes[0,0].set_ylabel("Count")

axes[0,1].pie(df['Gender'].value_counts(),labels=df['Gend er'].value_counts().index,autopct='%1.1f%

%',colors=['lightblue','lightcoral']) axes[0,1].set_title("Gender Distribution")
#

plot 3:Age distribution

axes[1,0].hist(df['Age'].dropna(),bins=8,color='lightgreen',

edgecolor='black') axes[1,0].set_title("Age Distribution") axes[1,0].set_xlabel("Age") axes[1,0].set_ylabel("Frequency")
#

plot 4:survival count

axes[1,1].bar(df['Survived'].value_counts().index,df['Survi

ved'].value_counts(),color=['lightblue','lightcoral'])

axes[1,1].set_title("Survival Count") axes[1,1].set_xlabel("Survived

(0 = No, 1 = Yes)") axes[1,1].set_ylabel("Count")

#

plot 5:fare vs age

axes[2,0].scatter(df['Age'],df['Fare'],color='orange',edgecolors='black') axes[2,0].set_title("Fare vs Age") axes[2,0].set_xlabel("Age") axes[2,0].set_ylabel("Fare") plt.tight_layout() plt.show() Output:

1. Histogram of Passenger information of Titanic Code:

#

write your code here for histogram import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

#
Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for

#

Write your code here for Histogram
#

write your code here for histogram
plt.hist(data['Age'],bins=30,edgecolor='k')
#

plt.title('Age Distribution') plt.xlabel('Age')

plt.ylabel('Frequency') plt.title('Age Distribution') plt.show()

Output:

1. Bar plot of survival rate of passengers Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin',

axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

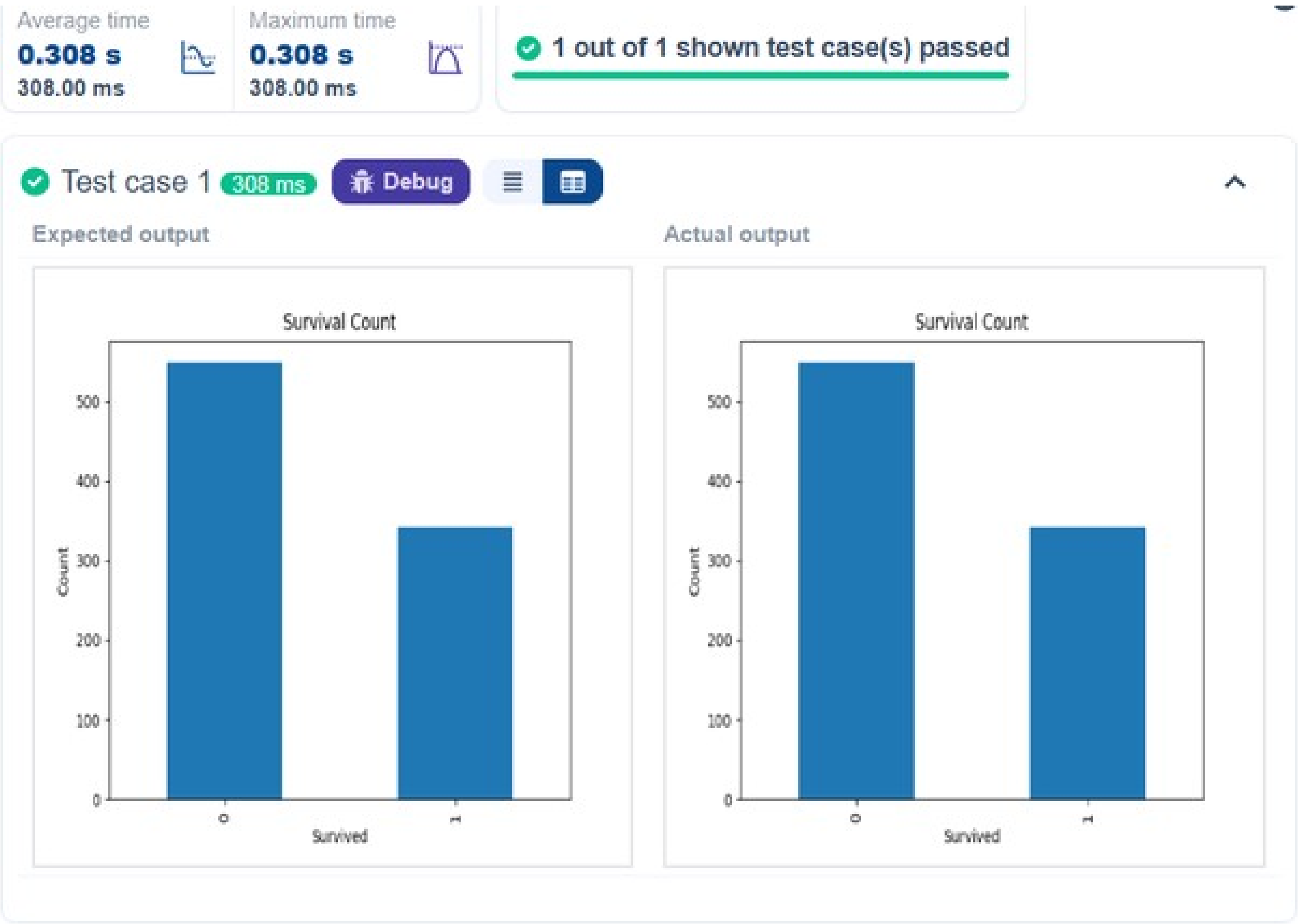
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Bar Plot for Survival Rate survival_counts = data['Survived'].value_counts() survival_counts.plot(kind='bar')

plt.title('Survival Count') plt.xlabel('Survived')

plt.ylabel('Count') plt.show() Output:



1. Bar plot for survival by Gender Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True) data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Bar Plot for Survival by Gender

grouped=data.groupby('Sex') ['Survived'].value_counts().unstack()

grouped.plot (kind='bar', stacked=True) plt.title('Survival by

Gender') plt.xlabel('Gender') plt.ylabel('Count') plt.legend(['Not

Survived','Survived']) plt.show()

Output:

1. Bar plot for survival by Pclass Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True) data.drop('Cabin', axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Bar Plot for Survival by Pclass

survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack()

survival_counts.plot(kind='bar',stacked=True) plt.title('Survival by Pclass') plt.ylabel('Count') plt.legend(['Not Survived','Survived']) plt.show()

Output

1. Bar plot for survival by Embarked Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(),

inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Bar Plot for Survival by Embarked

survival_counts = data.groupby('Embarked_Q')

['Survived'].value_counts().unstack()

survival_counts.plot(kind='bar',stacked=True) plt.title('Survival by Embarked') plt.xlabel('Embarked') plt.ylabel('Count') plt.legend(['Not Survived','Survived']) plt.show() Output:

1. Box plot for Age Distribution Code:

import pandas as pd
import matplotlib.pyplot as plt
#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(),

inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Box Plot for Age by Pclass data.boxplot(column='Age',by='Pclass')

plt.title('Age by Pclass') plt.xlabel('Pclass') plt.ylabel('Age') plt.suptitle("") plt.show()

Output:

1. Box plot for Age by Survived Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(),

inplace=True)

```
data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Box Plot for Age by Survived data.boxplot(column='Age',by='Survived')

plt.title('Age by Survival') plt.xlabel('Survived')

plt.ylabel('Age') plt.suptitle("") plt.show() Output:
```

1. Box plot for fare by Pclass Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(),

inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Box Plot for Fare by Pclass data.boxplot(column='Fare',by='Pclass')

plt.title('Fare by Pclass') plt.suptitle(') plt.xlabel('Pclass') plt.ylabel('Fare')

plt.show() Output:

1. Scatter plot for Age vs Fare Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(),
inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin',

axis=1, inplace=True)

#

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Box Plot for Fare by Pclass

plt.scatter(data['Age'],data['Fare']) plt.title("Age vs. Fare")

plt.xlabel("Age") plt.ylabel("Fare") plt.show()

Output:

1. Scatter plot for Age vs Fare by Survived Code:

import pandas as pd import

matplotlib.pyplot as plt

#

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning data['Age'].fillna(data['Age'].median(),

inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

#
Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

#

Write your code here for Scatter Plot for Age vs. Fare by Survived

colours=data['Survived'].map({0: 'red',1: 'blue'})

plt.scatter(data['Age'],data['Fare'],c=colours) plt.title("Age vs. Fare by Survival") plt.xlabel("Age") plt.ylabel('Fare') plt.show() Output: